AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT249
Practical modern statistics

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 4/Dec. /2025

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 90 Minutes

Instructions:

- Total Marks: **60**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part1: Q-1 [15 marks]

A- [10 marks] Answer the following 7 questions as True or False (1 mark),) and write the correct statement if the answer is false(1mark) :

1. ARMA (0, 0) model is the white noise model.
2. A time series X_t is stationary in mean if it has a zero mean, $E(X_t) = 0$.
3. The seasonal factors for a seasonal component of a time series with period T sum to one over the period.
4. The process of noise reduction in the non-seasonal additive model is called differencing.
5. The optimal choice for α is the value which minimizes the sum of squared errors (SSE).
6. Techniques such as differencing or normalization may be needed to induce stationarity time series.
7. The sample autocorrelation at lag k is calculated using the k pairs of points.

B- [5 marks] Fill in the blanks in the following 5 statements:

1- The _____ of a time series into its component parts, such as trend, seasonal, and irregular components, can help identify patterns in the data.
2- A _____ is a sequence of data points measured at regular time intervals.
3- The larger the standard deviation of W_t, the more the _____ obscures the trend, and the noisier the series is said to be.
4- If $\theta 1 = \cdots = \theta q = 0$, then the _____ term is said to be of order zero, and the model can be described as ARMA(p, 0).
5- _____ is the process of using a time series model to predict future values of the series based on past data.

Part2: Each question is worth 15 marks. Answer the following questions:

Q–1: [3+3+3+3+3 marks]:

The time plot for the time series of annual average temperatures in Central England is shown in Figure 1.

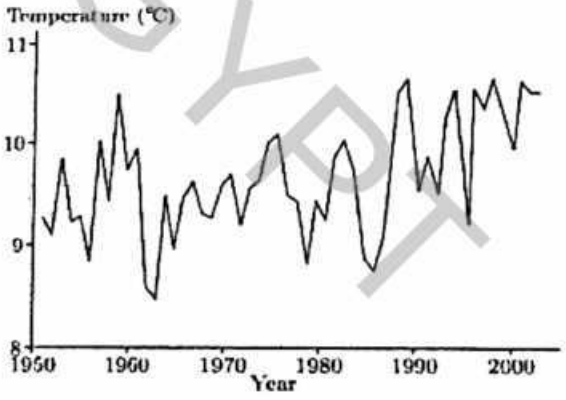


Figure 1

(a) Explain why this time series has no seasonal component.

(b) How important is the irregular component in this time series? Comment briefly on the impact that this has on visualizing the trend.

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Two transformations of the time plot of monthly sales by a company between January 1965 and May 1971 are shown in Figure 2. Figure 2(a) shows the time plot of the logarithms of the monthly sales. The time plot of the square roots of the monthly sales is shown in Figure 2(b).

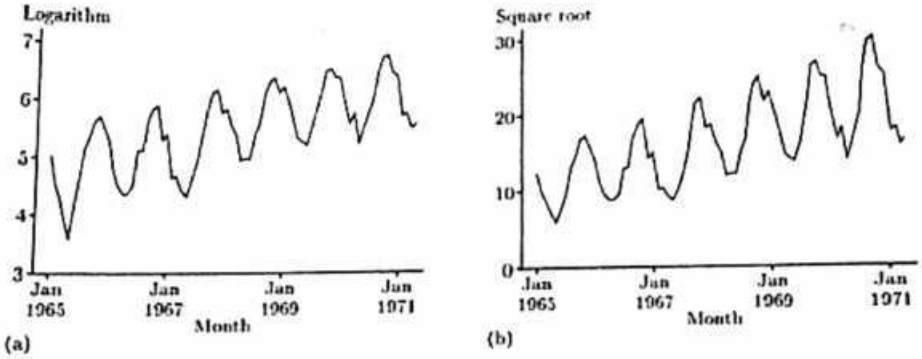


Figure 2

(c) In your opinion, which of the two-time series in Figure 2 would it be more appropriate to describe by an additive model?

(d) In any time series, the autocorrelation at lag 0 is equal to 1. Explain why this is so.

(e) Give an example of a sequence of numbers that represents the first few values of a seasonal component for a quarterly additive decomposition time series model.

Q–2: [3+3+3+6 marks]:

(a) How the order of the moving average should be chosen when smoothing a time series?

(b) Table 1 shows the first three values of a time series. Use simple exponential smoothing with $\alpha = 0.75$ and starting value $\hat{x}_1 = x_1$ to obtain the1-step ahead forecast of x_4.

Table 1

t	x_t
1	0.52
2	0.36
3	0.48

(c) Obtain the seasonal factor for Quarter 2, for the estimated seasonal factors for a quarterly time series shown below in Table 2.

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Table 2

Quarter	Seasonal factor
1	−1.1634
2	?
3	1.0654
4	0.5418

(d) Figure 3 shows the time plot of the natural logarithms of average house prices. The first 2 values of the time series are **10.83013** in January 1996 and **10.84458** in February 1996.

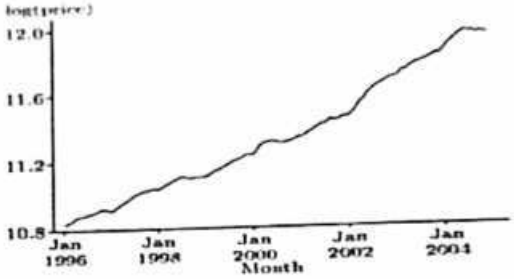


Figure 3

1- Why is Holt's exponential smoothing to be preferred over simple exponential smoothing for this time series?
2- Suggest appropriate initial values for the level & the slope.

Q–3: [2+2+2+2+3+4 marks]:
Temperatures of a chemical process (in °C) were measured at hourly intervals. Figure 4(a) shows the time plot of the temperatures, and Figures 4(b) and 4(c) show time plots of the first and second differences, respectively.

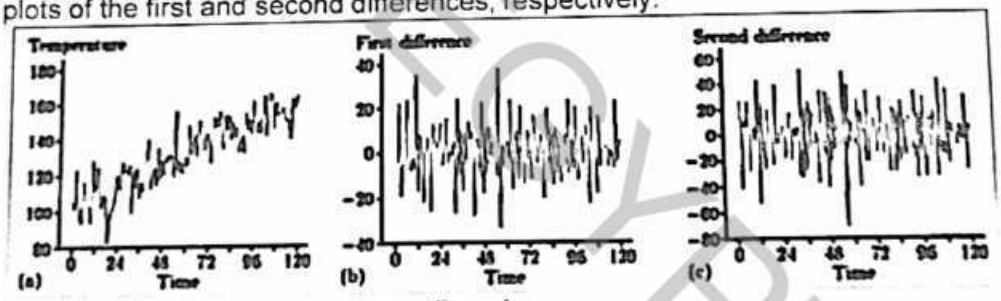


Figure 4

(a) Identify the order of differencing that appears to be required to obtain stationarity in mean. Justify your answer.
(b) Does the differenced series that you identified in part (a) appear to be stationary in variance? Justify your answer.

The correlogram Figure 5(a) and the partial correlogram Figure 5(b) for a stationary time series are shown below in Figure 5

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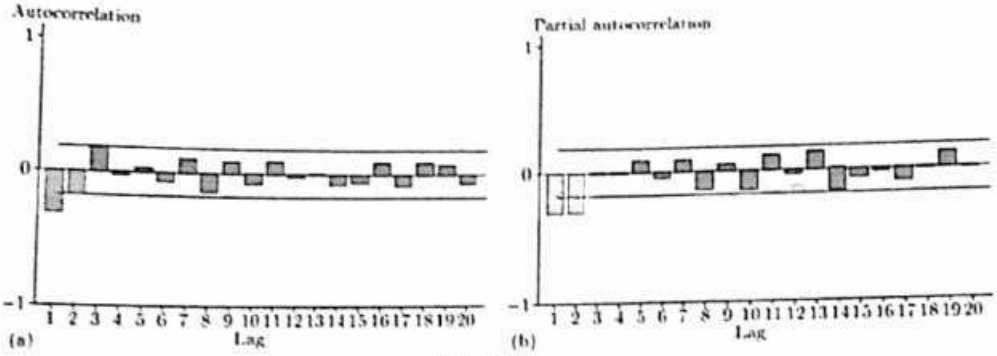


Figure 5

(c) Identify two plausible ARMA models for this time series. Justify your choices.

(d) Apply the principle of parsimony to select one or more 'best' candidate models.

(e) The series of the quarterly UK index of production differenced twice to obtain a stationary time series. Each of the following models is suitable for the twice-differenced series. Write down each model for the original series in the form ARIMA (p, d, q).

An MA(1) model. 2- An AR(2) model. 3- The white noise model.

(f) Let Y_t denote a non-seasonal time series, X_t a stationary series, and Z_t white noise. Classify the following models for Y_t as ARIMA (p, d, q) for suitable values of p, d, and q.

1. Model A: $X_t = 0.5X_{t-1} + Z_t - 0.2Z_{t-1}$ and $X_t = Y_t$.
2. Model B: $X_t = 0.05X_{t-1} - 0.4X_{t-2} + Z_t - 0.1Z_{t-1}$ and $X_t = Y_t - Y_{t-1}$.

End of questions

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